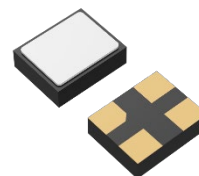


Description

The ABM14 series is an ultra-miniature AT-Cut MHz quartz crystal offered in a 1.0mm x 0.8mm x 0.33mm four-pad SMD package. Tight frequency accuracy and stability of ± 10 ppm over operating temperature range of -30°C to $+85^{\circ}\text{C}$, low plating load (CL) value of 8pF, and low Equivalent Series Resistance (ESR) is still achieved in the ultra-compact and thin package size. The ABM14 series offers industry standard frequencies (40MHz, 59.97MHz, and 76.8MHz), common for wearables, IoT, Bluetooth / Bluetooth Low Energy (BLE), and Ultra-Low Power MCU's/SoC's/Transceivers end applications.



Features

- Ultra-miniature At-Cut MHz Crystal (1.00 x 0.8 x 0.33mm package)
- Ideally suited for space constraint IoT, Wearables & Wireless applications
- Simultaneously optimized for low plating load & ESR over extended temperature range
- Enhanced performance for start-up time and power savings with Low Energy SoC's
- Low profile ideal for height constraint designs feature
- [REACH/RoHS II Compliant](#) | [MSL Level N/A](#)

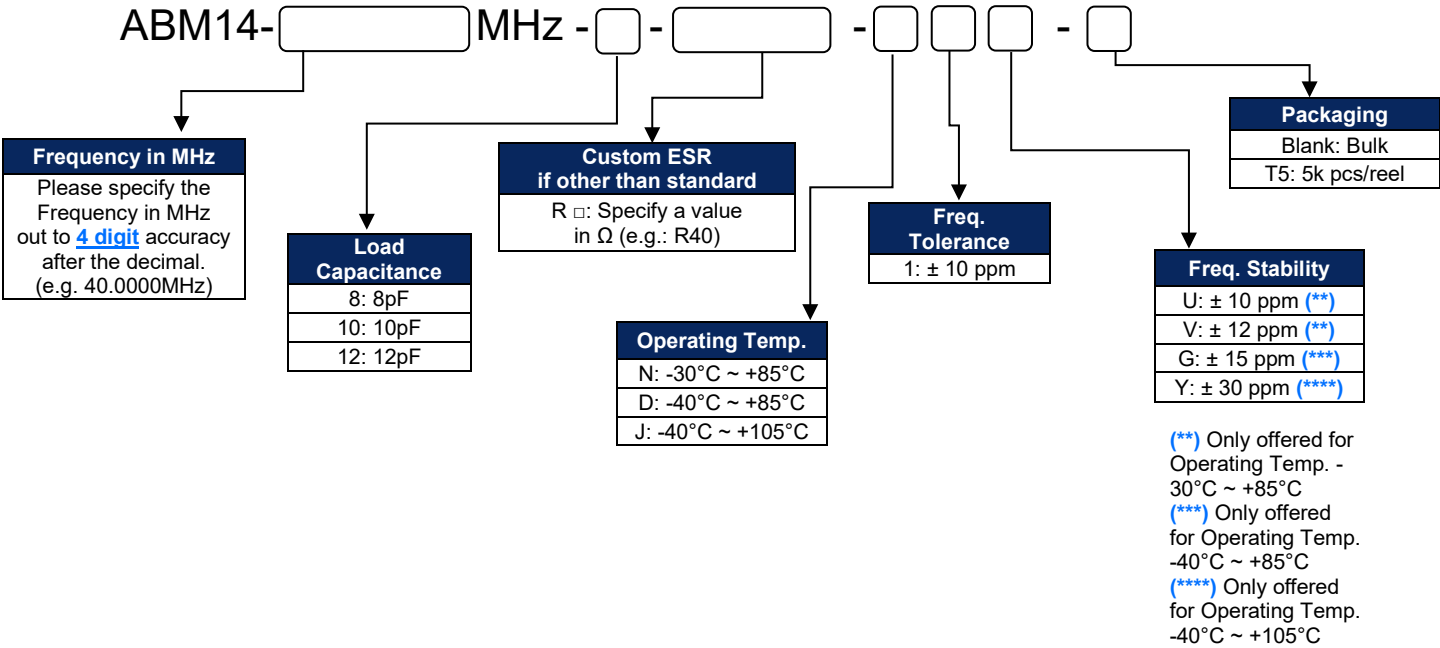
Typical Applications

- Wearables
- Wireless Modules
- Internet of Things (IoT)
- Bluetooth / Bluetooth Low Energy (BLE)
- Machine-to-Machine (M2M) Connectivity
- Ultra-Low Power MCU's, SoC's, Transceivers
- Near Field Communication
- ISM Band Applications

Electrical Specifications

Parameters	Min.	Typ.	Max.	Units	Notes
Frequency Range	37.4000		79.9600	MHz	
Standard Available Frequencies	40.0000, 59.9700, 76.8000			MHz	Contact Abracon for Nonstandard Frequencies
Operation Mode	Fundamental				
Operating Temperature Range	-30		+85	$^{\circ}\text{C}$	
Storage Temperature Range	-40		+85	$^{\circ}\text{C}$	
Frequency Tolerance @ $+25^{\circ}\text{C}$	-10		+10	ppm	See Options
Frequency Stability over the Operating Temperature Range (ref. to $+25^{\circ}\text{C}$)	-10		+10	ppm	See Options
Equivalent Series Resistance (R1)			60	Ω	40.0000 – 60.0000MHz
			30	Ω	76.8000 – 80.0000MHz
Load Capacitance (CL)		8.0		pF	See Options
Drive Level		10	100	μW	
Aging (1 year) @ $+25^{\circ}\text{C}$	-1		+1	ppm	40.0000 – 60.0000MHz
	-3		+3	ppm	76.8000 – 80.0000MHz
Insulation Resistance	500			M Ω	@ 100Vdc $\pm$ 15V

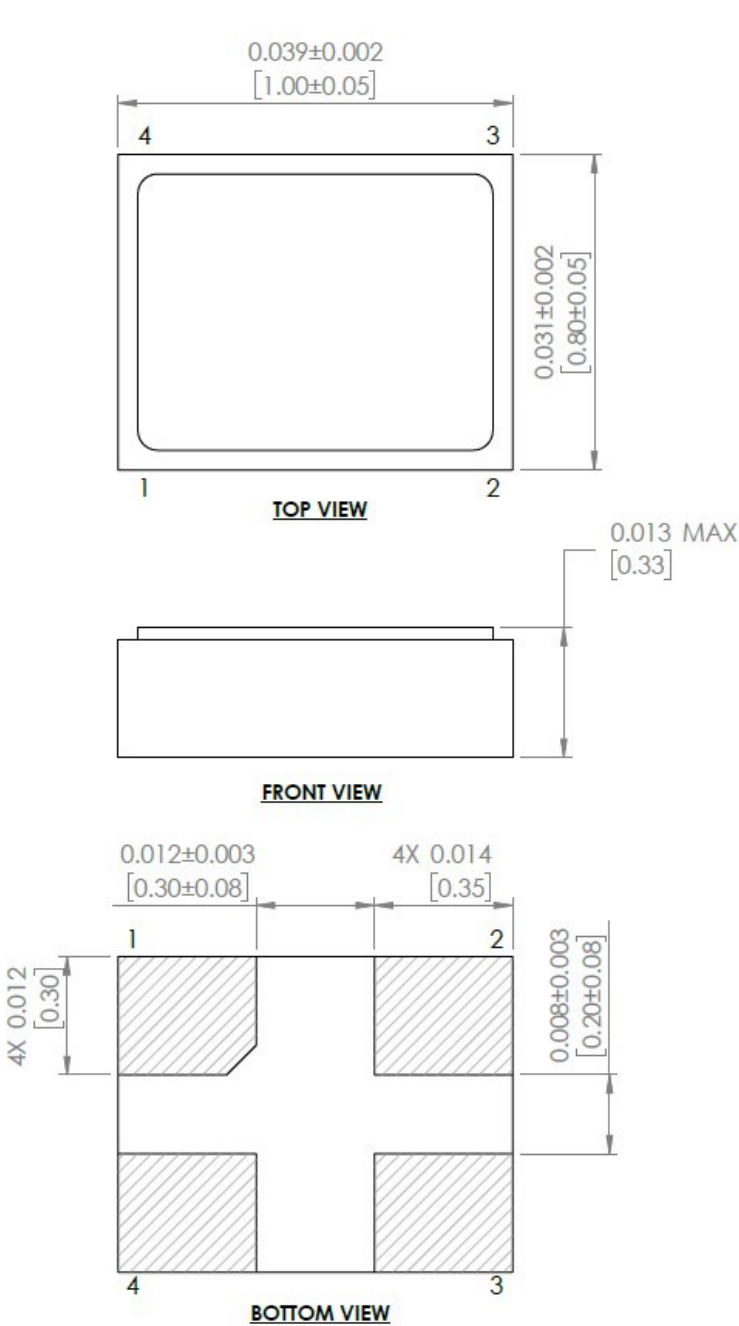
Part Identification [\[Note 1\]](#)



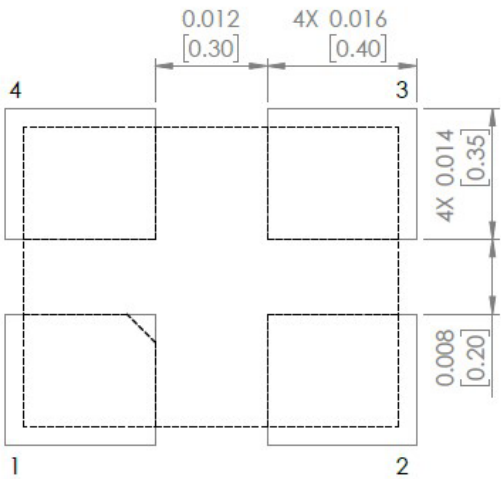
Note 1: Contact Abracon for part number requests with carrier frequency callouts up to 5 & 6 digit accuracy after the decimal.

Contact ABRACON for tighter Frequency Stability options.

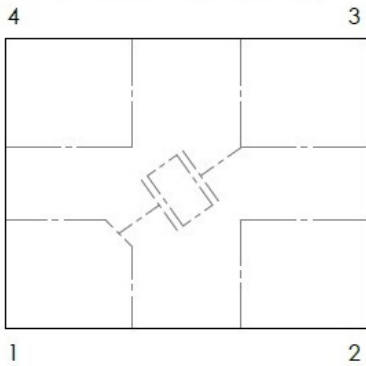
Mechanical Dimensions



RECOMMENDED LAND PATTERN



INTERNAL CONNECTION



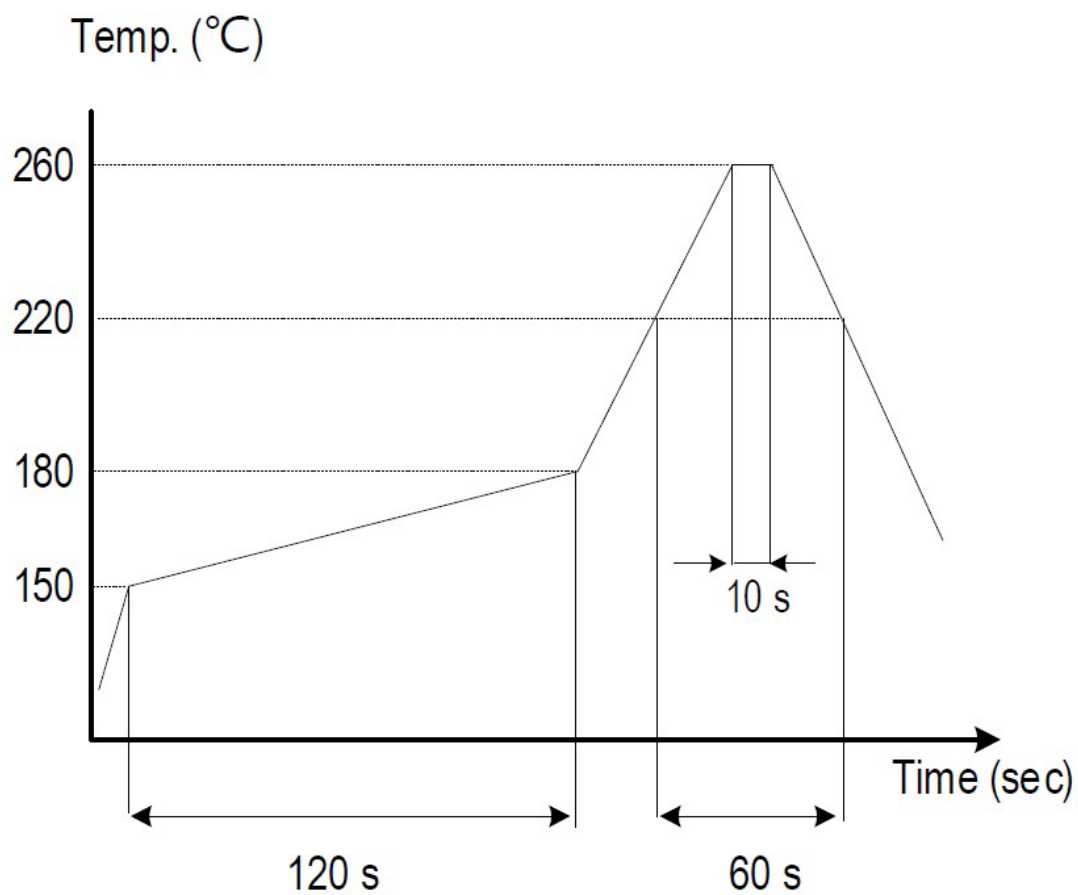
Pin #	Function
1	XTAL
2	GND (Connection with cover)
3	XTAL
4	NC (No internal connection, leave floating or connect to GND)

Dimensions: inches [mm]

Recommended Reflow Profile

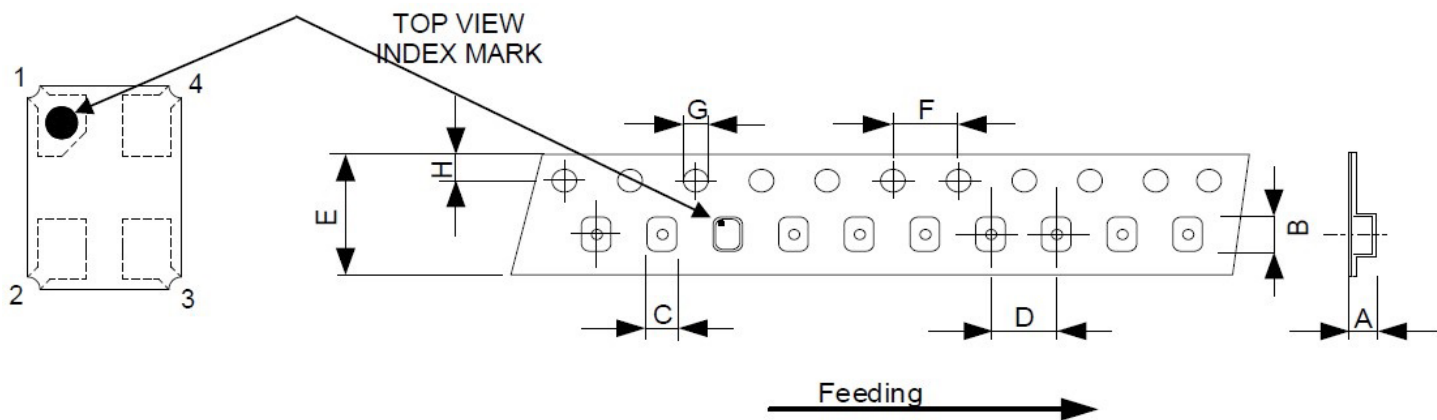
Total time: 200 sec. Max.

Solder melting point: 220 °C

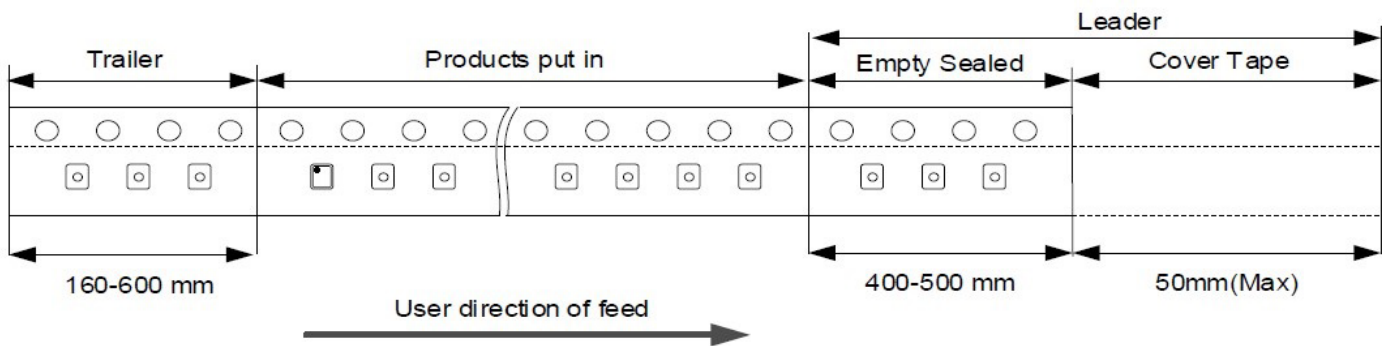


Packaging

T5: Tape and reel (5,000pcs/reel)

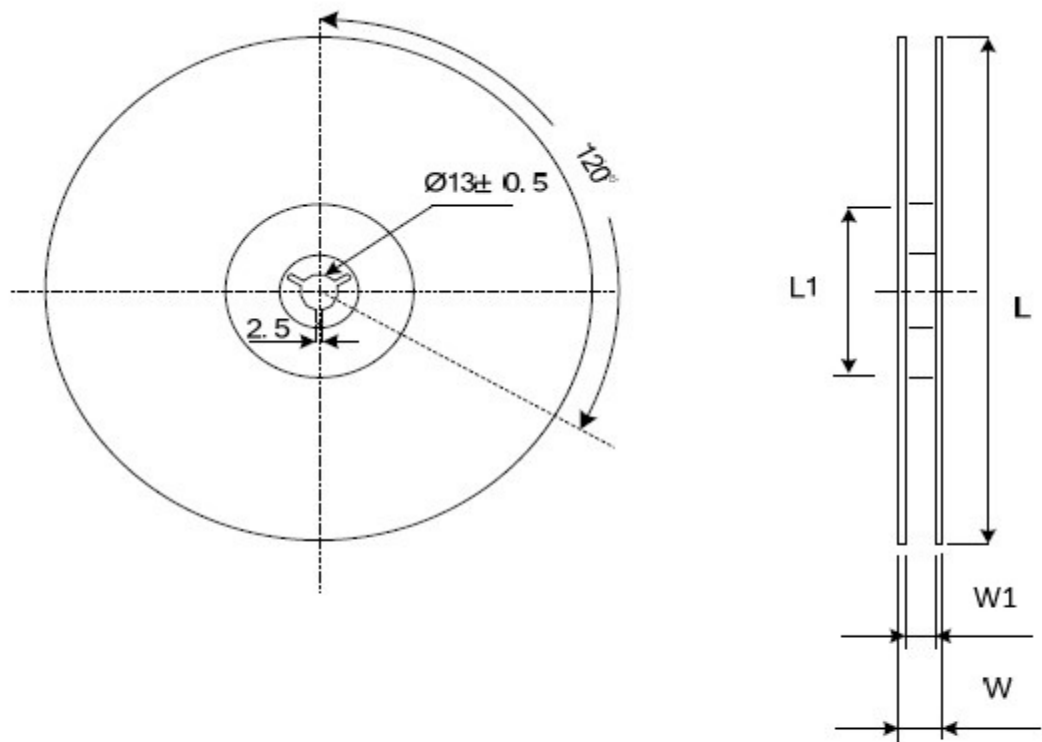


A	B	C	D	E	F	G	H
0.45	1.15	0.95	4.00	8.00	4.00	1.55	1.75
±0.05	±0.05	±0.05	±0.10	±0.20	±0.10	±0.05	±0.10



Dimensions: mm

Packaging continued



L	L1	W	W1
178	60.2	11.5	8
±1.00	±0.50	±0.2	+1/-0

Dimensions: mm